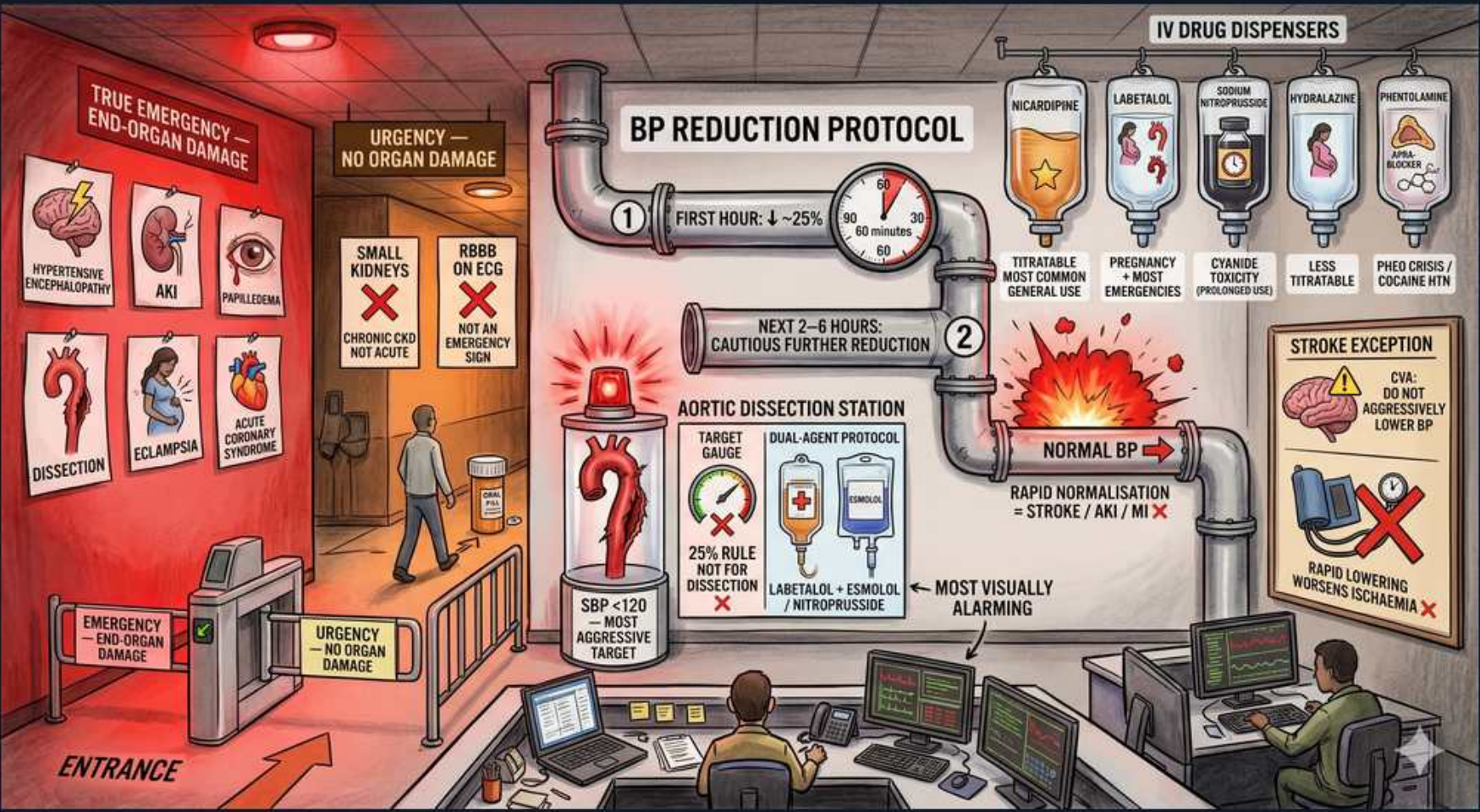


Hypertension — Emergency vs Urgency & BP-Lowering Protocol



1. Emergency = end-organ damage

WHAT YOU SEE

Red poster: TRUE EMERGENCY = END-ORGAN DAMAGE.

WHAT IT ENCODES

Hypertensive emergency = severely elevated BP (usually >180/120) WITH acute target-organ damage. It requires admission (usually ICU), IV titratable therapy, and arterial monitoring. The defining feature is the organ injury, not the BP number — a patient at 250/140 with no acute damage is an urgency, while one at 185/115 with papilledema or encephalopathy is an emergency. Goal: controlled, gradual reduction, not normalization.

BOARD TRAP

Treating a BP of 220/120 with IV antihypertensives because the number is alarming, even though the patient is asymptomatic with normal labs and exam — WRONG. The discriminator: it is the presence of acute end-organ damage, not the absolute number, that defines an emergency and justifies IV therapy; without damage this is urgency, managed with oral agents.

2. End-organ damage gallery

WHAT YOU SEE

Icons: hypertensive encephalopathy, AKI, papilledema, eclampsia, dissection, acute coronary syndrome.

WHAT IT ENCODES

Acute target-organ damage of hypertensive emergency: brain (hypertensive encephalopathy, ischemic/hemorrhagic stroke), eye (papilledema, retinal hemorrhages — grade III/IV retinopathy), kidney (acute kidney injury, hematuria), heart (ACS, acute pulmonary edema/HF), aorta (acute dissection), and pregnancy (eclampsia/severe preeclampsia). Each scenario dictates a specific drug and target.

BOARD TRAP

Calling chronically stable findings like long-standing LVH or baseline proteinuria 'acute end-organ damage' — WRONG. The discriminator: emergency requires NEW, ACUTE organ injury (rising creatinine, papilledema, encephalopathy, ACS, dissection, eclampsia), not chronic sequelae of hypertension.

3. Urgency = no organ damage

WHAT YOU SEE

Banner: URGENCY = NO ORGAN DAMAGE.

WHAT IT ENCODES

Hypertensive urgency = severely elevated BP (often >180/120) with NO acute end-organ damage. Manage by restarting/optimizing ORAL agents and reducing BP gradually over 24-48 hours in the outpatient setting, with close follow-up. Rapid IV lowering is harmful here. Common cause is medication nonadherence.

BOARD TRAP

Admitting a hypertensive-urgency patient to the ICU for rapid IV BP reduction to normal — WRONG and dangerous (causes hypoperfusion). The discriminator: with no acute organ damage, use oral agents and gradual outpatient lowering; reserve IV therapy and rapid reduction for true emergencies.

4. Small kidneys / RBBB not emergencies

WHAT YOU SEE

Notes: SMALL KIDNEYS = chronic CKD not acute; RBBB on ECG NOT an emergency sign.

WHAT IT ENCODES

Distractor findings that are CHRONIC, not acute end-organ damage: bilaterally small kidneys (chronic CKD/long-standing HTN, not acute injury) and an isolated right bundle branch block (a baseline conduction pattern, not acute ischemia). Neither converts an urgency into an emergency.

BOARD TRAP

Diagnosing hypertensive emergency because imaging shows small kidneys or the ECG shows RBBB — WRONG. The discriminator: these are chronic/baseline findings; an emergency needs evidence of ACUTE injury (rising creatinine, ST changes/ACS, encephalopathy), so an asymptomatic patient with these findings is an urgency.

5. First hour: reduce <=25%

WHAT YOU SEE

Pipe gauge: FIRST HOUR reduce ~25%.

WHAT IT ENCODES

In MOST hypertensive emergencies, lower the mean arterial pressure (MAP) by no more than ~10-25% during the FIRST hour using IV titratable agents. This protects autoregulation in brain, heart, and kidney. Exceptions where you go faster/lower: aortic dissection (SBP <120 rapidly) and severe preeclampsia/eclampsia. Exception where you go slower/higher: acute ischemic stroke.

BOARD TRAP

Rapidly normalizing BP to 120/80 in the first hour of a hypertensive emergency — WRONG. The discriminator: dropping BP below the autoregulatory range causes watershed ischemia (stroke, MI, AKI); limit the first-hour reduction to ~25% of MAP except in aortic dissection and eclampsia.

6. Next 2-6 hours cautious

WHAT YOU SEE

Step 2: NEXT 2-6 HOURS, CAUTIOUS FURTHER REDUCTION.

WHAT IT ENCODES

After the initial ~25% first-hour reduction, lower BP to approximately 160/100-110 over the next 2-6 hours, then cautiously toward normal over the following 24-48 hours if the patient is stable. This staged approach prevents hypoperfusion while still reversing the acute damage.

BOARD TRAP

Achieving full normalization within the first 6 hours of an emergency — WRONG. The discriminator: the protocol is deliberately staged (25% in hour 1, ~160/100 by 2-6h, normal over 24-48h); compressing the timeline risks ischemic organ injury.

7. Aortic dissection station

WHAT YOU SEE

Aorta diagram: TARGET SBP <120, DUAL-AGENT, labetalol/esmolol + nitroprusside, MOST AGGRESSIVE.

WHAT IT ENCODES

Acute aortic dissection is the most aggressive target: rapidly lower SBP to <120 mmHg AND heart rate to <60 bpm within ~20 minutes to reduce aortic wall shear stress (dP/dt). Give a BETA-BLOCKER FIRST (IV esmolol or labetalol) to control HR, THEN add a vasodilator (nitroprusside or nicardipine). Pain control with opioids also helps.

BOARD TRAP

Starting nitroprusside (a vasodilator) before beta-blockade in aortic dissection — WRONG. The discriminator: a vasodilator alone triggers reflex tachycardia that INCREASES shear stress and propagates the dissection; control heart rate with a beta-blocker first. Also, the ~25% rule does NOT apply here — dissection needs aggressive SBP <120.

8. IV drug dispensers

WHAT YOU SEE

IV bags: nicardipine, labetalol, sodium nitroprusside, hydralazine, phentolamine.

WHAT IT ENCODES

Titratable IV agents for emergencies, matched to scenario: nicardipine/clevidipine (most general use, neuro-friendly), labetalol/esmolol (pregnancy, dissection, most settings), nitroprusside (potent, short-acting; cyanide risk), nitroglycerin (ACS/flash pulmonary edema), hydralazine (pregnancy), and phentolamine (catecholamine excess — pheochromocytoma, cocaine/MAOI crisis).

BOARD TRAP

Using a single default agent for every emergency regardless of the organ involved — WRONG. The discriminator: the agent is scenario-specific — labetalol for eclampsia, beta-blocker-first then vasodilator for dissection, nitroglycerin for ACS/pulmonary edema, phentolamine for pheochromocytoma/cocaine.

9. Nicardipine = most general use

WHAT YOU SEE

Tag: NICARDIPINE, titratable, most common general use.

WHAT IT ENCODES

Nicardipine is a titratable IV dihydropyridine CCB and a versatile first choice for most hypertensive emergencies, including hypertensive encephalopathy and stroke-related HTN (it does not raise intracranial pressure or worsen cerebral autoregulation the way nitroprusside can). Clevidipine is an ultra-short-acting alternative.

BOARD TRAP

Defaulting to nitroprusside for a neuro emergency when nicardipine is available — WRONG. The discriminator: nitroprusside can raise intracranial pressure and risks cyanide toxicity; nicardipine/clevidipine are preferred, controllable choices for most emergencies including neurologic ones.

10. Labetalol = pregnancy

WHAT YOU SEE

Tag: LABETALOL, pregnancy, most emergencies.

WHAT IT ENCODES

Labetalol is a combined alpha-1 and beta blocker — preferred in pregnancy/eclampsia (alongside IV hydralazine and oral nifedipine) and safe across most emergencies, including aortic dissection (HR + BP control in one agent). Its mixed blockade avoids reflex tachycardia. Avoid in decompensated HFrEF, bradycardia, severe asthma, and cocaine toxicity (relative — unopposed alpha concern).

BOARD TRAP

Choosing an ACEi, ARB, or nitroprusside for a hypertensive emergency in a pregnant patient — WRONG. The discriminator: ACEi/ARBs are teratogenic and nitroprusside risks fetal cyanide accumulation; labetalol (or hydralazine/nifedipine) is the safe choice for hypertensive emergency in pregnancy.

11. Nitroprusside = cyanide risk

WHAT YOU SEE

Tag: SODIUM NITROPRUSSIDE, cyanide, prolonged use.

WHAT IT ENCODES

Sodium nitroprusside is a potent arterial AND venous vasodilator with rapid on/off control, but its metabolism releases cyanide and thiocyanate — risking toxicity with high doses, prolonged use (>24-48h), or renal/hepatic impairment. It can also raise intracranial pressure and cause coronary steal. Watch for unexplained metabolic acidosis and altered mental status (cyanide).

BOARD TRAP

Continuing a nitroprusside infusion for days in a patient with renal failure and attributing new confusion/lactic acidosis to the underlying illness — WRONG. The discriminator: suspect cyanide/thiocyanate toxicity; limit duration/dose, avoid in renal failure, and choose nicardipine or labetalol when prolonged therapy is anticipated.

12. Rapid normalization harms

WHAT YOU SEE

Explosion: RAPID NORMALISATION -> STROKE / AKI / MI X.

WHAT IT ENCODES

Over-rapid BP lowering drops perfusion below the shifted autoregulatory threshold in chronically hypertensive patients, causing watershed ischemia: ischemic stroke, AKI, and MI. This is the core rationale for the staged 25%-first-hour protocol. The goal is controlled reduction to a safe range, never abrupt normalization.

BOARD TRAP

Believing that the faster BP is brought to normal, the safer the patient — WRONG, it is the opposite. The discriminator: rapid normalization causes ischemic organ injury; chronically hypertensive brains/kidneys autoregulate at higher pressures, so reduce gradually.

13. Stroke exception (CVA)

WHAT YOU SEE

Brain panel: STROKE EXCEPTION, CVA, DO NOT aggressively lower BP; lowering worsens ischemia.

WHAT IT ENCODES

Acute ISCHEMIC stroke is the permissive-hypertension exception: do NOT lower BP unless it exceeds 220/120 (then reduce ~15% cautiously), OR the patient is a thrombolysis/thrombectomy candidate, in which case lower to <185/110 before tPA and keep <180/105 for 24h afterward. Higher BP maintains perfusion of the ischemic penumbra. (Acute hemorrhagic stroke has its own, lower targets, e.g. SBP ~140.)

BOARD TRAP

Aggressively normalizing BP of 200/110 in a patient with an acute ischemic stroke who is not getting tPA — WRONG. The discriminator: permissive hypertension preserves penumbral perfusion; only treat if >220/120 (no tPA) or >185/110 (before tPA). Do not apply the 25% emergency rule to acute ischemic stroke.